AI Sycophancy and AI Psychosis in Financial Technology

FTEC5660 HW3 Reflection

The central risk

AI systems become dangerous when fluent agreement is mistaken for independent judgment. AI sycophancy describes a model's tendency to accommodate a user's stated beliefs, preferences or desired conclusion, even when the evidence does not support them. The phrase AI psychosis is less precise. It is not a formal clinical diagnosis, but it is often used to describe situations in which prolonged interaction with an AI reinforces implausible beliefs or weakens a user's ability to distinguish generated narrative from verified reality. The two risks can form a feedback loop: the user offers a preferred interpretation, the model validates it, and that validation gives the next prompt greater certainty. In financial technology, this loop can influence real investments, borrowing decisions and risk assessments, so politeness and fluency cannot substitute for evidence.

How sycophancy enters financial analysis

A financial question often arrives with a hidden conclusion. An investor may ask why a stock is undervalued rather than whether it is undervalued. A founder may ask an AI to justify an aggressive revenue forecast instead of testing the assumptions behind it. If the model treats the wording of the question as an instruction to agree, it can select supportive facts, ignore contradictory evidence and produce a confident answer that feels like independent confirmation. The output may contain accurate individual figures while the overall conclusion remains biased because the model decided which figures deserved attention.

This matters because many financial decisions already suffer from confirmation bias. An agreeable model lowers the effort required to find arguments for an existing view. It can also create an illusion of consensus: one person's assumption returns in the voice of a seemingly separate analyst. Repeating the question in several forms does not create independent evidence when the same model is adapting to the same user. A useful financial agent should therefore be allowed to disagree, identify missing data and return an inconclusive answer when the evidence is insufficient.

How repeated agreement can distort belief

The concern described as AI psychosis goes beyond a single incorrect answer. In a long conversation, the model remembers the user's framing and continues the story built in earlier turns. Because the model produces coherent language, speculative statements can gradually acquire the appearance of established facts. If a user interprets that coherence as consciousness, personal loyalty or privileged insight, each new response may seem to confirm a relationship or pattern that exists only within the conversation. The model does not need an intention to deceive for the interaction to become harmful; prediction, personalization and repeated affirmation are enough to make an unsupported narrative feel stable.

Financial markets provide especially fertile material for this process because future prices are uncertain and almost any outcome can be wrapped in a plausible story. A user who believes an AI has discovered a secret market signal may interpret ordinary price movements as confirmation and losses as proof that the market is hiding the truth. The financial consequence can be immediate, but the deeper problem is epistemic: the user has lost a reliable method for deciding what counts as evidence. When a conversation starts shaping major financial or personal decisions, independent human judgment and external records become essential.

What the homework agent taught me

My undergraduate and postgraduate training in financial technology has emphasized models, data and automation. This homework made me think more carefully about where judgment should remain outside the

language model. In my Fisher agent, the model does not decide whether a company passes the three investment criteria. The agent retrieves the permitted data, SQLite performs the financial calculations, and Python compares each metric with its industry median. DeepSeek receives the verified verdicts and values only after those steps, and its role is limited to writing a concise explanation. If the model call fails, the system returns a deterministic summary instead of changing the result.

This design reduces sycophancy because a user's preferred conclusion cannot directly alter the arithmetic or threshold. It also reduces the chance that a fluent explanation will be mistaken for the source of the verdict. However, deterministic rules are not automatically objective. I still chose the indicators, dates and comparison benchmarks. A fixed rule can encode a poor assumption, and an industry median can be an unsuitable standard for a particular business. The lesson is therefore not to remove human judgment, but to make its location visible and reviewable.

Practices I would carry into FinTech work

I would use the following practices when an AI system contributes to financial analysis or advice:

I would begin with a neutral question before asking the model to defend a preferred position. I would require disconfirming evidence and record the source supporting each material claim. I would also separate retrieved facts, calculations, interpretation and uncertainty in the output, while keeping financial arithmetic and policy thresholds in testable code rather than free-form prose. For high-impact or emotionally charged decisions, I would consult an independent person and pause the AI interaction whenever it begins to replace external verification.

The most important change in my own thinking is that an AI system should be evaluated not only by whether it produces a correct answer, but also by whether its process helps the user preserve contact with evidence. A system that challenges weak premises, marks uncertainty and exposes its sources may feel less agreeable, yet it is more useful for financial decision-making. For me, responsible agent design now means building both a path to an answer and a path for the user to question that answer.